

Part 2

Hypothesis Testing: Example

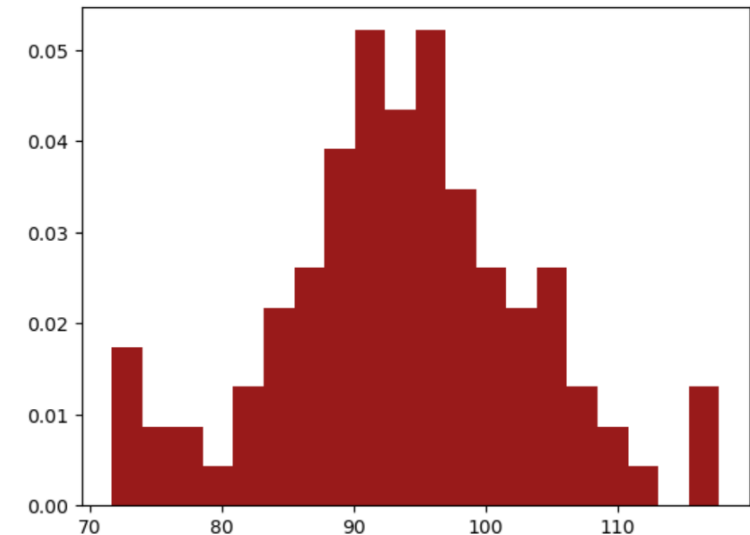
HYPOTHESIS TESTING. WORKING EXAMPLE

A call center claim that the average waiting time for customers is at most 90 seconds .
You are not so sure of this claim, and want to challenge this fact with data.
You collect some data, by taken a sample of 100 randomly chosen calls.

The data evidence that you obtain is the following:

- a data sample that seems normal (by its histogram)
- a sample mean = 93 seconds
- a sample standard deviation = 10 seconds.

With this information, can you reject the call center claim about customer waiting time?

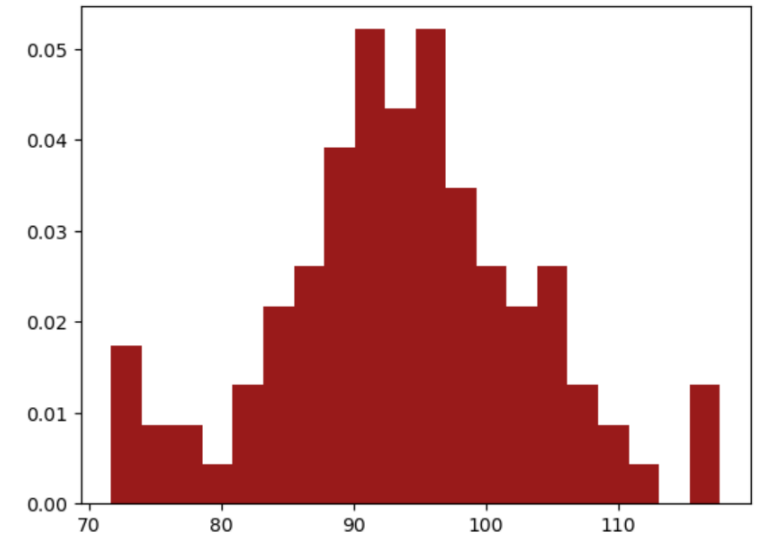


HYPOTHESIS TESTING. WORKING EXAMPLE

Some facts we can deduce/assume already:

1- Because sample histogram looks normally distributed (at least for a sample of that size), we will assume that the population is normally distributed.

2- Because The population is normally distributed and the sample size is big enough ($100 \gg 30$), then we will assume that the sample standard deviation and the population standard deviation are virtually equal. So, $\sigma = 10\text{secs}$

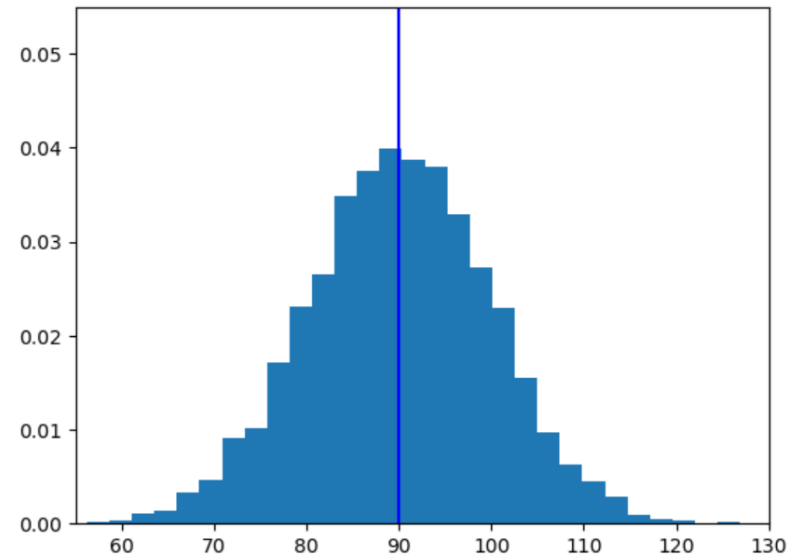


HYPOTHESIS TESTING. WORKING EXAMPLE

What we have.

The call center claim that waiting times are according to the following distribution.

Normal distribution
Mean = 90s
Standard deviation = 10s

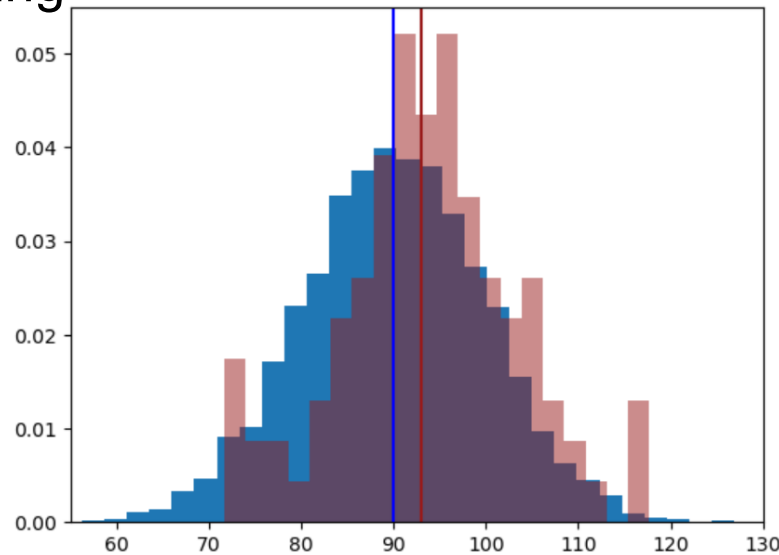


HYPOTHESIS TESTING. WORKING EXAMPLE

What we have.

The call center claim that waiting times are according to the following distribution.

Normal distribution
Mean = 90s
Standard deviation = 10s



Our 100 size sample follows the red distribution, which is a bit more to the right.

Normal-ish distribution
Mean = 93s
Standard deviation = 10s

This graph is suspicious, but does not prove anything!!

We need to do our Hypothesis testing work

HYPOTHESIS TESTING. WORKING EXAMPLE

So, we rephrase the problem as a hypothesis testing problem

A sample of $N=100$ is drawn from a population which is normally distributed and has a known standard deviation of 10. the sample mean is 93, and the claim to test is that the population mean is 90 or less.

- a) Write the Null and Alternative Hypothesis
- b) Is it a one sided or a two-sided test?
- c) Can we reject the Null Hypothesis
- d) Should I hire this call center to handle my phone call?

HYPOTHESIS TESTING. WORKING EXAMPLE

So, we rephrase the problem as an hypothesis testing problem

A sample of $N=100$ is drawn from a population which is normally distributed and has a known standard deviation of 10. the sample mean is 93, and the claim to test is that the population mean is 90 or less.

- a) Write the Null and Alternative Hypothesis
- b) Is it a one sided or a two sided test?
- c) Can we reject the Null Hypothesis with significance level $\alpha = 1\%$?
- d) Should I hire this call center to handle my phone call?

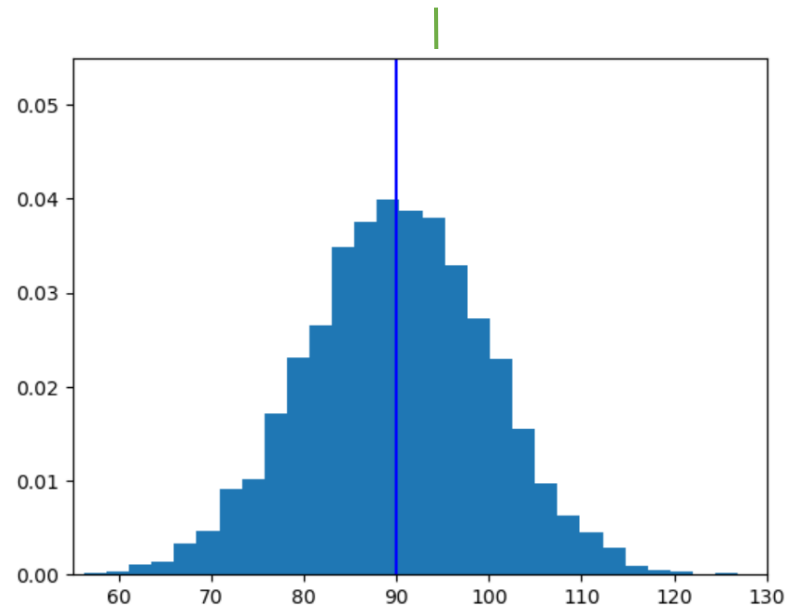
*If this is not given, I need to chose myself
significance level BEFORE doing the test.*



Let's solve this problem

HYPOTHESIS TESTING. WORKING EXAMPLE

The call center claim that waiting times are according to the following distribution and the mean is 90 seconds OR LESS.



Population $\mu = 90$
 $\sigma = 10$

a) Write the Null and Alternative Hypothesis

$$H_0: \mu \leq 90$$

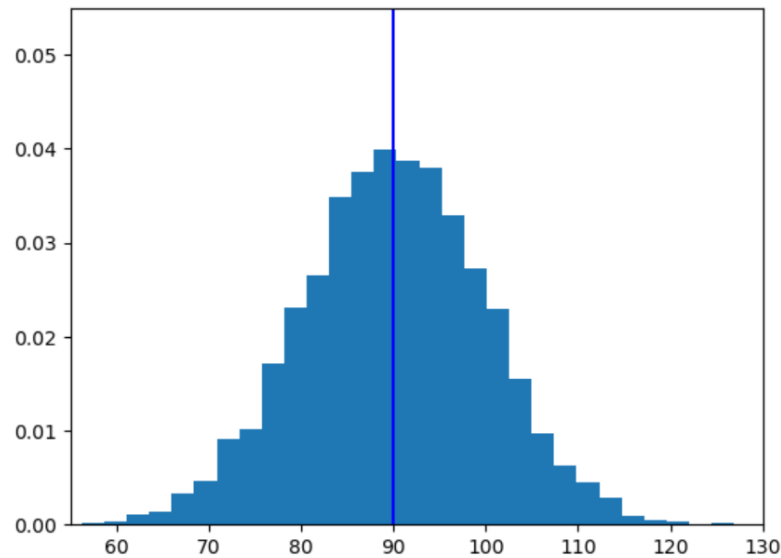
$$H_1: \mu > 90$$

b)) Is it a one sided or a two sided test?

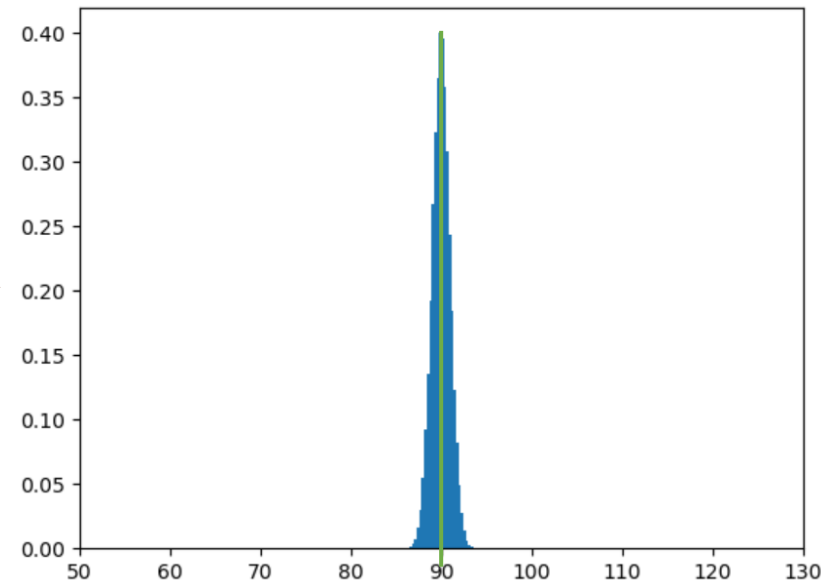
One sided

HYPOTHESIS TESTING. WORKING EXAMPLE

If the call center claim is true, how likely is that I get an average value of 93s for a sample of size $N = 100 \Rightarrow$ find the sampling distribution and the standard error, (se)



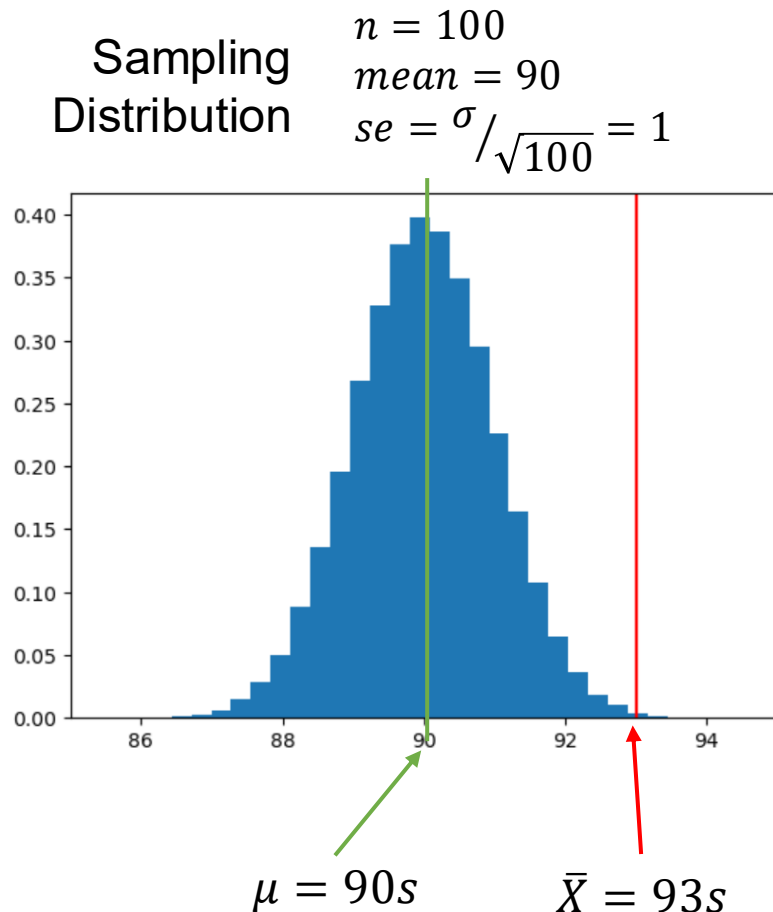
Population $\mu = 90$
 $\sigma = 10$



Sampling Distribution $n = 100$
 $mean = 90$
 $se = \sigma / \sqrt{100} = 1$

HYPOTHESIS TESTING. WORKING EXAMPLE

Let's zoom in in the SAMPLING DISTRIBUTION



Let's measure the distance from $\bar{X}$ to μ in terms of how many standard errors separate them.

Let's call this 'distance' the Z value

$$Z = \frac{\bar{X} - \mu}{se} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{93 - 90}{1} = 3$$

A sample mean of 93 seconds, is 3 standard errors to the right of the 'expected' mean.

Is that a lot? => **it depends on my significance level!!**

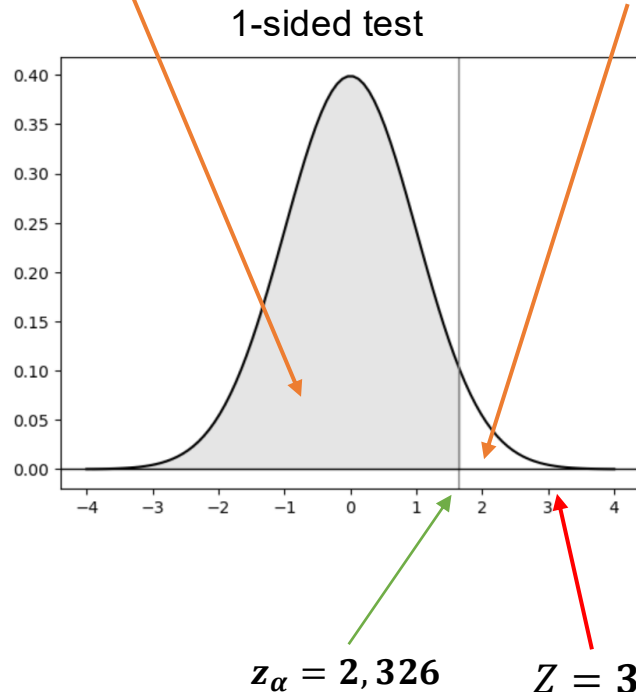
HYPOTHESIS TESTING. WORKING EXAMPLE

Is the sample mean 'far' from the expected mean?

The right question to ask: Is the value $Z = 3$ in my acceptance region or in my rejection region?

Acceptance region

Rejection region



- The rejection region is obtained by selecting a confidence level.
- For $\alpha = 0.01$, the critical $z_{\alpha} = 2.326$

```
import scipy.stats
alpha=0.01
z_right= scipy.stats.norm.ppf(1-alpha)
print(z_right)
```

2.3263478740408408

- Our value Z is inside the rejection region.
- **We should then reject the Null hypothesis that the waiting call has an average of 90 seconds!**
- Think twice before hiring that call center

HYPOTHESIS TESTING. WORKING EXAMPLE

Let's solve the complete example following the full procedure

Statement of the problem (short version)

Call center claim $\mu \leq 90$ seconds, and $\sigma = 10$. Normal distribution assumption ok.
I measure sample $N=100$ and find average $\bar{X} = 93$ second. Should I hire this call center?

Step 1. Set null hypothesis and type of test.

$$H_0: \mu \leq 90$$

$$H_1: \mu > 90$$

One side test. I reject hypothesis (don't hire center) if waiting time clearly $>90 \Rightarrow$ Rejection region to the right.

Step 2. Fix a confidence level.

Let's take $\alpha = 0.01$ (small number because I want to be confident that I do not commit a type 1 error).

Step 3. Compute critical level

Using python (or memory as this is a common alpha value) $z_\alpha = 2,326$

Step 4. Compute the Z-value

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{N}}} = 3 > 2.326 = z_\alpha$$



Reject Null Hypothesis
Don't Hire that Call Center

SOME USEFUL CRITICAL VALUES

Type of test	$\alpha = 0.1$	$\alpha = 0.05$	$\alpha = 0.01$
One sided. Rejection region at right $H_0: \mu \leq \mu_0$ $H_1: \mu > \mu_0$	1,282	1,645	2,326
One sided. Rejection region at left $H_0: \mu \geq \mu_0$ $H_1: \mu < \mu_0$	-1,282	-1,645	-2,326
Two sided. Rejection regions both extremes $H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$	-1,645 +1,645	-1,96 +1,96	-2,576 +2,576

*Calculated using this code.
Extend the table by computing your own significance levels!*

```
import scipy.stats
alpha=.05

# One sided test. Right Side
z_critic = scipy.stats.norm.ppf(1-alpha)

#One sided test; Left Side
z_critic = scipy.stats.norm.ppf(alpha)

#Two sided
z_left = scipy.stats.norm.ppf(alpha/2)
z_right = scipy.stats.norm.ppf(1- alpha/2)
```